

Stochastic Process

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Outline

- ① Stochastic Process
- ② Markov Process

Stochastic Process

- A stochastic process $\{X(t) : t \in T\}$ is a family of random variables indexed by a parameter t , which runs over an index set T .
- The parameter t usually denotes time.
- For any specific time t , $X(t)$ is a random variable.
- The index set T is called the parameter set.
- If T is countable, $\{X(t)\}$ is a discrete time stochastic process. If T is an interval, finite or infinite, $\{X(t)\}$ is a continuous time stochastic process.
- The set of possible values of $X(t)$ at any time t is called the state space, S .

Example

A business office has five telephone lines and any number of these lines may be in use at any time. The telephone lines are observed at regular intervals of 2 minutes.

X : Number of lines in use in every 2 minutes.

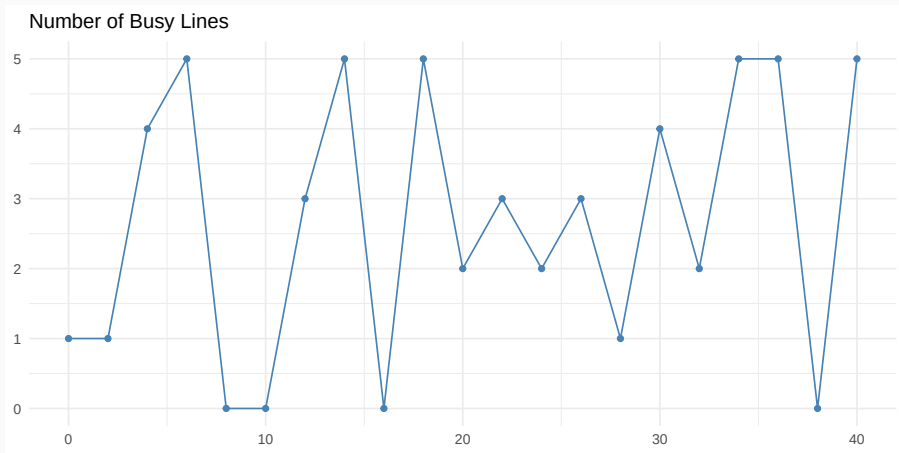
For $T = 0, 1, 2, 3, \dots$,

$$X(t) : \{X(0), X(1), X(2), \dots\} \quad \text{e.g. } X(t) : \{3, 2, 5, 0, 1, 0, 3, \dots\}$$

Here, $\{X(t) : t \in T\}$ is a stochastic process with parameter space $T = \{0, 1, 2, 3, \dots\}$ and state space $S = \{X : 0, 1, 2, 3, 4, 5\}$.

Graphical Example

An observation of this process might look like below.



Examples of some stochastic processes

1. Random walk model: Let X_t be a random variable that can be any of the two states [up (+1) or down (−1)] at time t .

X_t	+1	−1
probability	p	$1 - p$

Then, the process $R_t = R_{t-1} + X_t$ is called a random walk model.

2. Counting process: A stochastic process $\{N(t); t > 0\}$ is a counting process if $N(t)$ represents the total number of events that have occurred in time t .

3. Birth & death process: A stochastic process $\{N(t); t \geq 0\}$ with states $\{n = 0, 1, 2, \dots\}$ for which transition from state n may go only to either of the states $(n - 1)$ or $(n + 1)$ is a birth and death process.

Autocorrelation or Serial Correlation

- In statistics, the autocorrelation of a random process describes the correlation between values of the process at different times, as a function of the two times or of the time lag.
- Let X be some repeatable process, and i be some point in time after the start of that process. Then X_i is the value (or realization) produced by a given run of the process at time i .
- The autocorrelation between times s and t is defined as $R(s, t) = \text{Corr}(X_t, X_s)$, the correlation coefficient between X_t and X_s .

Markov Process

Markov Process

- Consider a finite or countably infinite set of points $(t_0, t_1, \dots, t_n, t)$, where $t_0 < t_1 < \dots < t_n < t$ and $t, t_i \in T$ ($i = 0, 1, 2, \dots, n$); T is a parameter space of the process $\{X(t)\}$.
- The dependence exhibited by the process $\{X(t) : t \in T\}$ is called Markovian dependence if the conditional distribution of $X(t)$ for a given value of $X(t_0), X(t_1), X(t_2), \dots, X(t_n)$ depends only on $X(t_n)$, which is the most recent known value of the process, i.e., if

$$\begin{aligned} \Pr[X(t) = x \mid X(t_n) = x_n, X(t_{n-1}) = x_{n-1}, \dots, X(t_0) = x_0] \\ = \Pr[X(t) = x \mid X(t_n) = x_n] \end{aligned}$$

The stochastic process exhibiting this property (Markov property) is called a Markov process.

Types of Markov Process

Parameter space		State space	
		Discrete	Continuous
Discrete		Markov chain	Discrete parameter, continuous MP
Continuous		Continuous parameter, discrete MC	Continuous parameter, continuous MP

Notations:

p_i = Probability that the process is in state i at a given time

$p_{ij}^{(n)}$ = n -step transition probability of state j from state i
= $\Pr[\text{process moves from } i \text{ to } j \text{ in } n \text{ steps}]$

Transition Probability Matrix (TPM)

A matrix containing probabilities of transition from one state to another. If there are k finite states of a Markov process, i.e. $S = \{1, 2, \dots, k\}$, the one-step TPM is

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1k} \\ p_{21} & p_{22} & \cdots & p_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1} & p_{k2} & \cdots & p_{kk} \end{pmatrix}$$

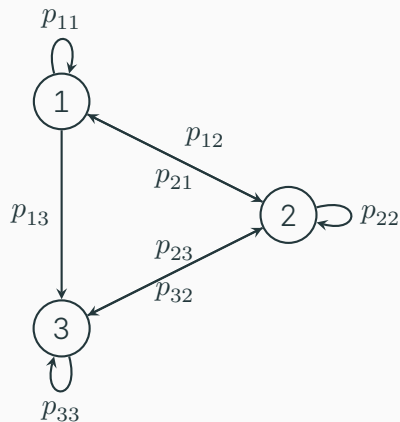
such that (i) $p_{ij} \geq 0$, $\forall i, j \in S$; (ii) $\sum_j p_{ij} = 1$ (row total 1).

Markov Process: Example 1

X = living status. State space
 $S = \{1, 2, 3\}$; where 1 = healthy,
2 = sick, 3 = dead.

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix}$$

Here, $p_{33} = 1$ and all other probabilities are less than 1.



Time-homogeneous Markov chains (or stationary Markov chains) are processes where

$$\Pr[X_{t+1} = x \mid X_t = y] = \Pr[X_t = x \mid X_{t-1} = y]$$

for all t . The probability of the transition is independent of t .

Classification of states

- **Accessible state:** A state j is said to be accessible from a state i (written $i \rightarrow j$) if a system started in state i has a non-zero probability ($p_{ij} > 0$) of transitioning into state j at some point.
- **Communicating states:** A state i is said to communicate with state j (written $i \leftrightarrow j$) if both $i \rightarrow j$ and $j \rightarrow i$. Here, $p_{ij} > 0$ and $p_{ji} > 0$.
- **Absorbing state:** A state i is called absorbing if it is impossible to leave this state. Therefore, state i is absorbing if and only if $p_{ii} = 1$ and $p_{ij} = 0$ for $i \neq j$.

Classification of states (cont.)

- **Transient state & recurrent state:** A state i is said to be transient if, given that we start in state i , there is a non-zero probability that we will never return to i . State i is recurrent (or persistent) if it is not transient. Recurrent states are guaranteed (with probability 1) to have a finite hitting time.
- **Periodic state & aperiodic state:** A state i has period k if any return to state i must occur in multiples of k time steps. A state is aperiodic if returns to state i can occur at irregular times ($k = 1$).

Multi-step transitions

- The n -step TPM is obtained by multiplying the one-step TPM by itself n times: $P^{(n)} = P^n$. Its (i, j) th entry, $p_{ij}^{(n)}$, is the probability of moving from state i to state j in n steps.
- If $A = (p_1, p_2, \dots, p_k)$ is the probability vector (distribution over states) at time 0, then the distribution after n steps is

$$A^{(n)} = A \cdot P^n$$

- This can be computed either as $A^{(n)} = A^{(n-1)} \cdot P$ (applying P one step at a time) or directly as $A \cdot P^{(n)}$ once $P^{(n)}$ is known — both give the same result.

Example 2

There are two telephone lines in an office and any number of these lines may be in use at any given time. Telephone lines are observed at regular intervals of 2 minutes. The initial probability vector of the states is

$$A = (0.3, 0.5, 0.2)$$

and the one-step transition probability matrix is

$$P = \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}$$

Assuming a time-homogeneous Markov chain X_t , determine the probabilities that no line, 1 line, and 2 lines are being used at (i) $t = 1$, (ii) $t = 2$ (starting time $t = 0$). Also find $\Pr[X_0 = 1, X_1 = 0, X_2 = 2]$.

Let 0: no line in use, 1: 1 line in use, 2: 2 lines in use, so $S = \{0, 1, 2\}$.

Example — solution

For $t = 1$:

$$A^{(1)} = AP = (0.3, 0.5, 0.2) \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix} = (0.23, 0.51, 0.26)$$

For $t = 2$:

$$A^{(2)} = A^{(1)}P = (0.23, 0.51, 0.26) \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix} = (0.225, 0.497, 0.278)$$

Example — solution

Find $\Pr[X_0 = 1, X_1 = 0, X_2 = 2]$.

$$\begin{aligned}\Pr[X_0 = 1, X_1 = 0, X_2 = 2] &= \Pr[X_2 = 2 \mid X_1 = 0] \Pr[X_1 = 0 \mid X_0 = 1] \Pr[X_0 = 1] \\ &= p_{02} p_{10} p_1\end{aligned}$$

using the Markov property, $\Pr[X_2 = 2 \mid X_1 = 0, X_0 = 1] = \Pr[X_2 = 2 \mid X_1 = 0]$.
So,

$$\Pr[X_0 = 1, X_1 = 0, X_2 = 2] = p_1 p_{10} p_{02} = 0.5 \times 0.3 \times 0.2 = 0.03$$

Steady State Probabilities

If, for an irreducible Markov chain (only one class, so all states communicate), all states are aperiodic and positive recurrent (returns in finite time), the distribution $A^{(n)} = A \cdot P^n$ converges as $n \rightarrow \infty$, and the limiting distribution is independent of the initial probabilities A :

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \lim_{n \rightarrow \infty} a_j^{(n)} = p_j$$

These are called steady state probabilities, and satisfy

$$p_j > 0, \quad \sum_{j=0}^m p_j = 1, \quad p_j = \sum_{i=0}^m p_i p_{ij}, \quad j = 0, 1, \dots, m$$

Since there are $m + 2$ equations but only $m + 1$ unknowns, one equation is redundant — use m of the $m + 1$ equations above together with $\sum_j p_j = 1$.

Example 3

Find the steady state probabilities for the Markov chain in Example 2.

At steady state,

$$(p_0, p_1, p_2) = (p_0, p_1, p_2) \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{pmatrix}$$

which gives the system

$$-0.8p_0 + 0.3p_1 + 0.1p_2 = 0$$

$$0.6p_0 - 0.5p_1 + 0.4p_2 = 0$$

$$p_0 + p_1 + p_2 = 1$$

(the third original balance equation is redundant given $\sum p_j = 1$).

Example — solution

Solving the system by elimination gives

$$p_0 = \frac{17}{77}, \quad p_1 = \frac{38}{77}, \quad p_2 = \frac{22}{77}$$

These are the long-run (steady state) probabilities that 0, 1, and 2 telephone lines are in use, respectively, regardless of the initial distribution A .